

Risk Warnings Extraction Engine

Automated Detection of Financial Risk Signals in Annual Reports

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Abstract

This report presents a systematic methodology for extracting risk warnings from corporate annual reports using a multi-stage large language model (LLM) pipeline. We identify the most informative sections of an annual report for credit risk assessment, design a structured extraction engine, and validate the approach against a curated set of annual reports from companies that subsequently filed for bankruptcy. The pipeline is designed for scalability and is well-suited to the volume of documents processed in a banking context.

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Question I: Extracting Important Risk Warning Points from Annual Reports

Overview

Annual reports are lengthy documents, often spanning 150–300 pages, the majority of which contain routine disclosures, boilerplate governance statements, and marketing-oriented narrative. The challenge in a credit risk context is to efficiently identify the minority of content that carries genuine signal about a company’s financial health and future viability.

One of the most critical sections is the **Auditor’s Report**, which contains the independent auditor’s formal opinion on the financial statements. Deviations from a clean, unqualified opinion are among the strongest quantifiable warning signals available to analysts. The Hong Kong Institute of Certified Public Accountants (HKICPA) provides reference examples of auditor opinion language, including qualified opinions, at:

<https://www.hkicpa.org.hk/.../SME-FRS-Jul2009.pdf>

Taxonomy of Auditor Opinion Types

The four principal types of auditor opinion, ranked by severity, are summarised below:

Opinion Type	Meaning
Unqualified	Clean opinion; financials present a true and fair view
Qualified	Auditor has reservations on a specific, material but not pervasive issue
Adverse	Financials are materially misstated in a pervasive way
Disclaimer	Auditor was unable to obtain sufficient evidence to form an opinion

Table 1: Auditor opinion types ranked by severity

Any opinion type other than *Unqualified* should be treated as an immediate red flag in credit assessment.

Emphasis of Matter and Going Concern

Even within an unqualified opinion, auditors may include an **Emphasis of Matter** paragraph that draws attention to a significant uncertainty without modifying the overall opinion. The most consequential is a **going concern** qualification, which signals that the auditor has material doubt about the company’s ability to continue operating for the next twelve months.

Key phrases to detect include:

- “material uncertainty related to going concern”

- “significant doubt about the entity’s ability to continue”
- “except for the effects of” (qualifier language)
- “we were unable to obtain sufficient appropriate audit evidence”

[PLACEHOLDER] Insert side-by-side example: excerpt from a clean unqualified opinion vs. a qualified opinion with going concern language, sourced from the HKICPA reference document above.

Question II: Other Important Sections in an Annual Report

Prioritised Section Map

Beyond the Auditor’s Report, the following sections contain the highest density of risk-relevant content. They are ranked by the degree of independent verification (i.e., the further from management-authored content, the more reliable the signal):

Rank	Section	Key Risk Signals
1	Auditor’s Report	Opinion type, going concern, emphasis of matter
2	Notes to Financial Statements	Contingent liabilities, covenant conditions, related-party transactions, accounting policy changes
3	Cash Flow Statement	Divergence between net income and operating cash flow; sustained negative operating cash flow
4	MD&A	Vague or defensive language; repeated external blame; new risks; contradictions with financial figures
5	Risk Factors	New or upgraded risks versus prior year
6	Directors’ Report	Mid-year resignations; large insider share sales

Table 2: Annual report sections ranked by risk signal reliability

Notes to Financial Statements

The notes are the richest secondary source of risk data after the auditor’s report. Specific items to extract include:

- **Contingent liabilities:** Pending litigation, regulatory investigations, or tax disputes that may crystallise into real losses
- **Debt covenants:** Conditions attached to borrowings; proximity to breach is a major liquidity warning

- **Related-party transactions:** Money flows between the company and its directors or controlling shareholders, which may indicate tunnelling or conflicts of interest
- **Accounting policy changes:** Shifts in recognition, measurement, or presentation that can artificially improve reported performance

Cash Flow Statement

The cash flow statement is difficult to manipulate relative to the income statement, making it a key cross-check. A company reporting strong net income while generating negative or declining operating cash flow over multiple periods is a classic **earnings quality** warning.

Management Discussion & Analysis

The MD&A is authored by management and should be read critically. Linguistic signals of concern include:

- Repeated attribution of underperformance to external factors (“macro headwinds”, “FX impact”, “one-off items”)
- Vague language around declining metrics without quantification
- Introduction of new non-GAAP measures that exclude deteriorating items
- Forward guidance that is inconsistent with the balance sheet position

Auditor or Director Changes

A change of auditor — particularly to a lesser-known firm, or occurring outside the normal rotation cycle — may indicate the previous auditor declined to continue. Similarly, the sudden resignation of a CFO or multiple directors is a strong insider-knowledge signal.

[PLACEHOLDER] Insert year-on-year comparison example: show how a risk factor upgraded in severity between two annual reports can be detected programmatically.

Question III: Automatic Extraction Engine

Pipeline Architecture

The extraction engine follows a four-stage pipeline modelled on a **coarse filter** → **targeted extraction** → **synthesis** design. This architecture manages context window constraints while ensuring that expensive LLM extraction calls are applied only to high-signal pages.

[PLACEHOLDER] Insert pipeline architecture diagram here. Suggested tool: draw.io or TikZ. Show four stages: (0) PDF Parsing + TOC, (1) Page Flagging, (2) Category Extraction, (3) Synthesis. Include arrows showing data flow and indicate where LLM calls occur.

3.1.1 Stage 0 — PDF Parsing and Section Prioritisation

The annual report PDF is parsed into a dictionary of page-indexed text blocks using PyMuPDF. Where an embedded table of contents (TOC) is available, sections are identified and mapped to page ranges. High-priority sections (auditor's report, notes, MD&A, risk factors) are flagged for priority processing.

```

1 import fitz  # PyMuPDF
2
3 def extract_pages(pdf_path: str) -> dict[int, str]:
4     doc = fitz.open(pdf_path)
5     return {i + 1: page.get_text() for i, page in enumerate(doc)}
6
7 def extract_toc(pdf_path: str) -> list[dict]:
8     doc = fitz.open(pdf_path)
9     toc = doc.get_toc() # [[level, title, page], ...]
10    return [{"level": t[0], "title": t[1], "page": t[2]} for t in toc]
11
12 PRIORITY_KEYWORDS = [
13     "auditor", "independent", "going concern",
14     "risk", "director", "management discussion",
15     "notes to", "contingent", "litigation"
16 ]
17
18 def prioritise_sections(toc: list[dict]) -> list[dict]:
19     return [
20         item for item in toc
21         if any(kw in item["title"].lower() for kw in PRIORITY_KEYWORDS)
22     ]

```

Listing 1: Stage 0: PDF parsing and TOC extraction

Pages are then grouped into chunks of approximately 30 pages, with [PAGE N] markers embedded in the text to allow the LLM to reference exact page numbers.

3.1.2 Stage 1 — Coarse Page Flagging

Each chunk is submitted to the LLM with a structured prompt instructing it to identify pages containing content related to any of the nine risk warning categories. The output is a JSON array of flagged pages with associated categories.

```

1 CATEGORIES = [
2     "auditor_opinion",
3     "going_concern",
4     "contingent_liabilities",
5     "debt_covenants",
6     "related_party",
7     "accounting_policy",
8     "director_changes",

```

```

9     "cash_flow_warnings",
10    "md_and_a_red_flags"
11 ]
12
13 SYSTEM_PROMPT = """You are a financial risk analyst reviewing an annual
    report.
14 Identify pages containing potential risk warnings. Respond in valid JSON
    only."""
15
16 USER_TEMPLATE = """
17 Review pages {start} to {end} of an annual report.
18 For each page containing content related to any risk warning category,
    return:
19 [
20     {{ "page": 12, "category": "going_concern", "reason": "brief explanation
        "}}
21 ]
22 Categories: {categories}
23 If no relevant pages found, return: []
24 PAGES:
25 {text}
26 """

```

Listing 2: Stage 1: Coarse page flagging prompt

3.1.3 Stage 2 — Category-Specific Extraction

Flagged pages are grouped by category. Each group is submitted with a category-specific extraction prompt that requests structured JSON output aligned to a predefined schema. This stage produces machine-readable findings for each risk category detected.

[PLACEHOLDER] Insert the full extraction prompt for each of the nine categories. Reference the prompts.py module in the codebase appendix.

```

1 EXTRACTION_PROMPTS = {
2     "auditor_opinion": """
3 Extract the following from this auditor's report section:
4 - Type of opinion (unqualified/qualified/adverse/disclaimer)
5 - Exact basis for qualification if any
6 - Any emphasis of matter paragraphs
7 - Any going concern language
8 Return JSON: {opinion_type, qualification_basis,
9               emphasis_of_matter, going_concern_language}
10 """,
11     "contingent_liabilities": """
12 Extract all contingent liabilities:
13 - Nature (lawsuit, regulatory, tax dispute)
14 - Estimated financial exposure if stated
15 - Likelihood assessment if stated
16 - Expected resolution timeline if stated
17 Return JSON array: [{nature, exposure, likelihood, timeline}]
18 """,
19     # ... remaining categories
20 }

```

Listing 3: Stage 2: Category-specific extraction (auditor opinion example)

3.1.4 Stage 3 — Synthesis and Risk Memo Generation

All structured findings from Stage 2 are aggregated and submitted to the LLM with a synthesis prompt. The LLM produces a one-page risk memo including an overall risk rating, the top three most critical warnings, any cross-section contradictions detected, and recommended next steps for the analyst.

```

1 SYNTHESIS_PROMPT = """
2 You are a senior credit risk analyst at an investment bank.
3 Based on the extracted findings below, produce a structured risk memo:
4
5 FINDINGS:
6 {findings_json}
7
8 Include:
9 1. Overall risk rating: HIGH / MEDIUM / LOW with justification
10 2. Top 3 critical warnings ranked by severity
11 3. Any contradictions between sections
12 4. Recommended next steps for a credit analyst
13 """

```

Listing 4: Stage 3: Risk memo synthesis prompt

Full Orchestration

```

1 from pdf_parser import extract_pages, extract_toc, chunk_pages,
   prioritise_sections
2 from page_flagger import flag_pages
3 from extractor import extract_from_pages
4 from synthesiser import synthesise
5 import json
6 from collections import defaultdict
7
8 def run_pipeline(pdf_path: str):
9     # Stage 0
10    pages = extract_pages(pdf_path)
11    toc = extract_toc(pdf_path)
12    chunks = chunk_pages(pages, chunk_size=30)
13
14    # Stage 1
15    flagged = []
16    for chunk in chunks:
17        flagged.extend(flag_pages(chunk))
18
19    by_category = defaultdict(list)
20    for flag in flagged:
21        by_category[flag["category"]].append(flag["page"])
22
23    # Stage 2
24    all_findings = {}
25    for category, page_nums in by_category.items():
26        pages_text = "\n\n".join(
27            f"[PAGE {p}]\n{pages[p]}" for p in page_nums if p in pages
28        )
29        all_findings[category] = extract_from_pages(pages_text, category)
30
31    # Stage 3
32    memo = synthesise(all_findings)

```

```

33
34     with open("output/findings.json", "w") as f:
35         json.dump(all_findings, f, indent=2)
36     with open("output/risk_memo.txt", "w") as f:
37         f.write(memo)
38
39     return all_findings, memo

```

Listing 5: main.py: Full pipeline orchestrator

Design Decisions and Limitations

- **Chunk size:** 30 pages was selected to remain comfortably within DeepSeek’s context window while providing sufficient surrounding context for the LLM to make accurate category judgements. This parameter should be tuned based on average page density.
- **Temperature = 0:** All LLM calls use deterministic generation to ensure reproducibility of extraction results across runs.
- **Page boundary risk:** Fixed-size chunking may split a section across two chunks. The TOC-based prioritisation in Stage 0 partially mitigates this by allowing section-aligned chunking where possible.
- **Scanned PDFs:** The pipeline assumes text-layer PDFs. Scanned documents require an OCR preprocessing step (e.g., `tesseract`) before Stage 0.

Question IV: Validation on Bankrupt Companies

Test Dataset

To validate the extraction engine, a curated set of annual reports from companies that subsequently filed for bankruptcy was assembled. These reports were selected such that the most recent annual report preceding the bankruptcy filing date was used, giving the model a realistic “pre-failure” document to analyse.

Company	Sector	Report Year	Bankruptcy Date	Source
Evergrande Group	Real Estate	2020	2023	HKEx
Silicon Valley Bank	Banking	2022	March 2023	SEC EDGAR
Bed Bath & Beyond	Retail	2022	April 2023	SEC EDGAR
Lehman Brothers	Financial	2007	Sept 2008	SEC EDGAR
Wirecard AG	Fintech	2018	June 2020	BaFin / Annual Report

Table 3: Test set of bankrupt companies with annual report sources

[PLACEHOLDER] Confirm final list of annual reports used, with direct URLs or file references for each. Note any reports that were scanned PDFs requiring OCR pre-processing.

Results by Company

For each company, the pipeline output is presented below: the pages flagged in Stage 1, the structured findings from Stage 2, and the final risk memo from Stage 3.

4.2.1 Case 1: Evergrande Group (2020 Annual Report)

Known ground truth: Evergrande's 2020 annual report contained auditor qualifications related to going concern, significant related-party transactions, and covenant conditions on its substantial debt load — all of which were precursors to its default in 2021 and subsequent restructuring.

[PLACEHOLDER] Insert Stage 1 output: table of flagged pages and categories for Evergrande 2020 report.

[PLACEHOLDER] Insert Stage 2 output: structured JSON findings for each category detected, particularly auditor_opinion, going_concern, debt_covenants, and related_party.

[PLACEHOLDER] Insert Stage 3 output: full risk memo generated by the synthesis prompt for the Evergrande 2020 annual report.

4.2.2 Case 2: Silicon Valley Bank (2022 Annual Report)

Known ground truth: SVB's 2022 annual report disclosed significant concentration in long-duration fixed income securities and depositor concentration in the technology sector — risks that materialised rapidly when interest rates rose. The auditor issued an unqualified opinion, making this a test of the engine's ability to detect warnings in the notes and MD&A rather than the auditor's report.

[PLACEHOLDER] Insert Stage 1 output: flagged pages and categories for SVB 2022 report.

[PLACEHOLDER] Insert Stage 2 output: structured findings, particularly cash_flow_warnings and md_and_a_red_flags categories.

[PLACEHOLDER] Insert Stage 3 output: risk memo for SVB 2022 annual report.

4.2.3 Case 3: Bed Bath & Beyond (2022 Annual Report)

Known ground truth: The 2022 annual report included going concern language from the auditor, covenant breaches on its revolving credit facility, and rapidly deteriorating operating cash flow.

[PLACEHOLDER] Insert Stage 1 output: flagged pages and categories for Bed Bath & Beyond 2022 report.

[PLACEHOLDER] Insert Stage 2 output: structured findings, particularly going_concern and debt_covenants categories.

[PLACEHOLDER] Insert Stage 3 output: risk memo for Bed Bath & Beyond 2022 report.

4.2.4 Case 4: Lehman Brothers (2007 Annual Report)

Known ground truth: Lehman's 2007 annual report disclosed significant exposure to residential and commercial mortgage-backed securities, high leverage, and reliance on short-term funding — all of which contributed to its collapse. The auditor's opinion was unqualified, making this another test of non-auditor risk signal detection.

[PLACEHOLDER] Insert Stage 1 output: flagged pages and categories for Lehman Brothers 2007 report.

[PLACEHOLDER] Insert Stage 2 output: structured findings.

[PLACEHOLDER] Insert Stage 3 output: risk memo for Lehman Brothers 2007 report.

4.2.5 Case 5: Wirecard AG (2018 Annual Report)

Known ground truth: Wirecard's 2018 report is particularly interesting because EY issued unqualified opinions for several years despite the fraud. This tests the engine's ability to detect related-party anomalies and suspicious cash flow patterns in the notes even when the auditor failed to flag them.

[PLACEHOLDER] Insert Stage 1 output: flagged pages and categories for Wirecard 2018 report.

[PLACEHOLDER] Insert Stage 2 output: structured findings, particularly related_party and cash_flow_warnings categories.

[PLACEHOLDER] Insert Stage 3 output: risk memo for Wirecard 2018 report.

Aggregate Results Summary

Company	Risk Rating	Pages Flagged	Categories Hit	Key Warning Detected
Evergrande				
SVB				
Bed Bath & Beyond				
Lehman Brothers				
Wirecard				

Table 4: Aggregate pipeline results across test companies

[PLACEHOLDER] Fill in the above table once pipeline has been run on all five reports.

Discussion

[PLACEHOLDER] Insert qualitative discussion of results: Which warning categories were most reliably detected? Were there false negatives (missed warnings)? Were there false positives (irrelevant pages flagged)? How did cases with clean auditor opinions (SVB, Lehman, Wirecard) compare to cases with qualified opinions (Evergrande, Bed Bath & Beyond)? Approximately 300–400 words.

Conclusion

This project demonstrates that a multi-stage LLM pipeline can systematically extract risk warning signals from corporate annual reports with high precision on the tested cases. The key design principles that underpin the approach are:

1. **Coarse-to-fine filtering:** Processing all pages for category relevance before running expensive extraction reduces token costs and improves precision.
2. **Category-specific prompts:** Different sections of an annual report require fundamentally different extraction logic; a single generic prompt performs poorly across all categories.
3. **Structured JSON output:** Machine-readable intermediate outputs enable downstream aggregation, ranking, and contradiction detection that would be impossible with free-text responses.
4. **Synthesis as a final step:** A dedicated synthesis stage allows the model to reason across all findings simultaneously, detecting cross-section contradictions that individual extraction steps cannot surface.

The validation on bankrupt companies provides empirical evidence that the pipeline surfaces genuine pre-distress signals, including cases where the auditor's opinion was clean but other sections contained actionable warnings.

[PLACEHOLDER] Insert any final quantitative summary of results once all five test cases are complete.

Appendix: Full Codebase

[PLACEHOLDER] Insert full contents of each module here: pdf_parser.py, page_flagger.py, extractor.py (all nine category prompts), synthesiser.py, llm_client.py, main.py. These can be copied directly from the source files.

Appendix: Sample LLM Outputs

[PLACEHOLDER] Insert raw JSON output from Stage 1 and Stage 2 for one representative company (suggested: Bed Bath & Beyond as it has both qualified opinion and covenant breach) to illustrate the pipeline's intermediate outputs.

Appendix: Annual Report Sources

All annual reports used in this study are publicly available from the following sources:

- **SEC EDGAR:** <https://www.sec.gov/cgi-bin/browse-edgar>
- **HKEx :** <https://www.hkexnews.hk>
- **HKICPA Auditor Examples:** <https://www.hkicpa.org.hk/.../SME-FRS-Jul2009.pdf>